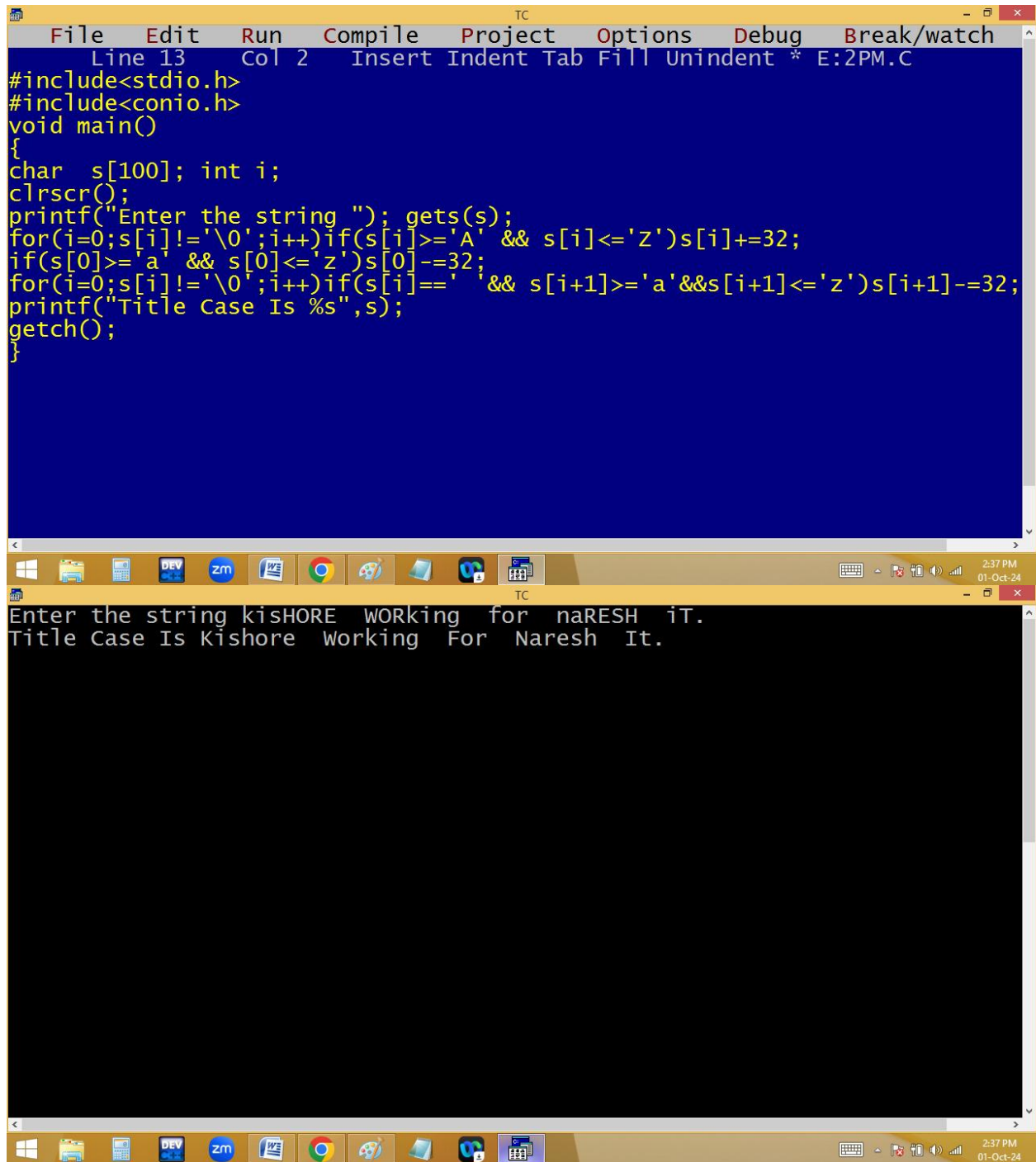


Capitalize Each Word:



The image shows a screenshot of a Turbo C++ (TC) IDE. The top window displays the source code for a program that converts a string to title case. The code uses the `gets()` function to read a string and then iterates through it, adjusting the ASCII values of uppercase and lowercase letters to achieve the title case format. The bottom window shows the program's execution, where the input string "kishORE WORKing for naRESH iT." is processed and the output "Title Case Is Kishore Working For Naresh It." is displayed. The Windows taskbar at the bottom indicates the system time as 2:37 PM on 01-Oct-24.

```
File Edit Run Compile Project Options Debug Break/watch
Line 13 Col 2 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s[100]; int i;
clrscr();
printf("Enter the string "); gets(s);
for(i=0;s[i]!='\0';i++)if(s[i]>='A' && s[i]<='Z')s[i]+=32;
if(s[0]>='a' && s[0]<='z')s[0]-=32;
for(i=0;s[i]!='\0';i++)if(s[i]==' ' && s[i+1]>='a'&&s[i+1]<='z')s[i+1]-=32;
printf("Title Case Is %s",s);
getch();
}
```

Enter the string kishORE WORKing for naRESH iT.
Title Case Is Kishore Working For Naresh It.

Title Case / Capitalize Each
Word:

s

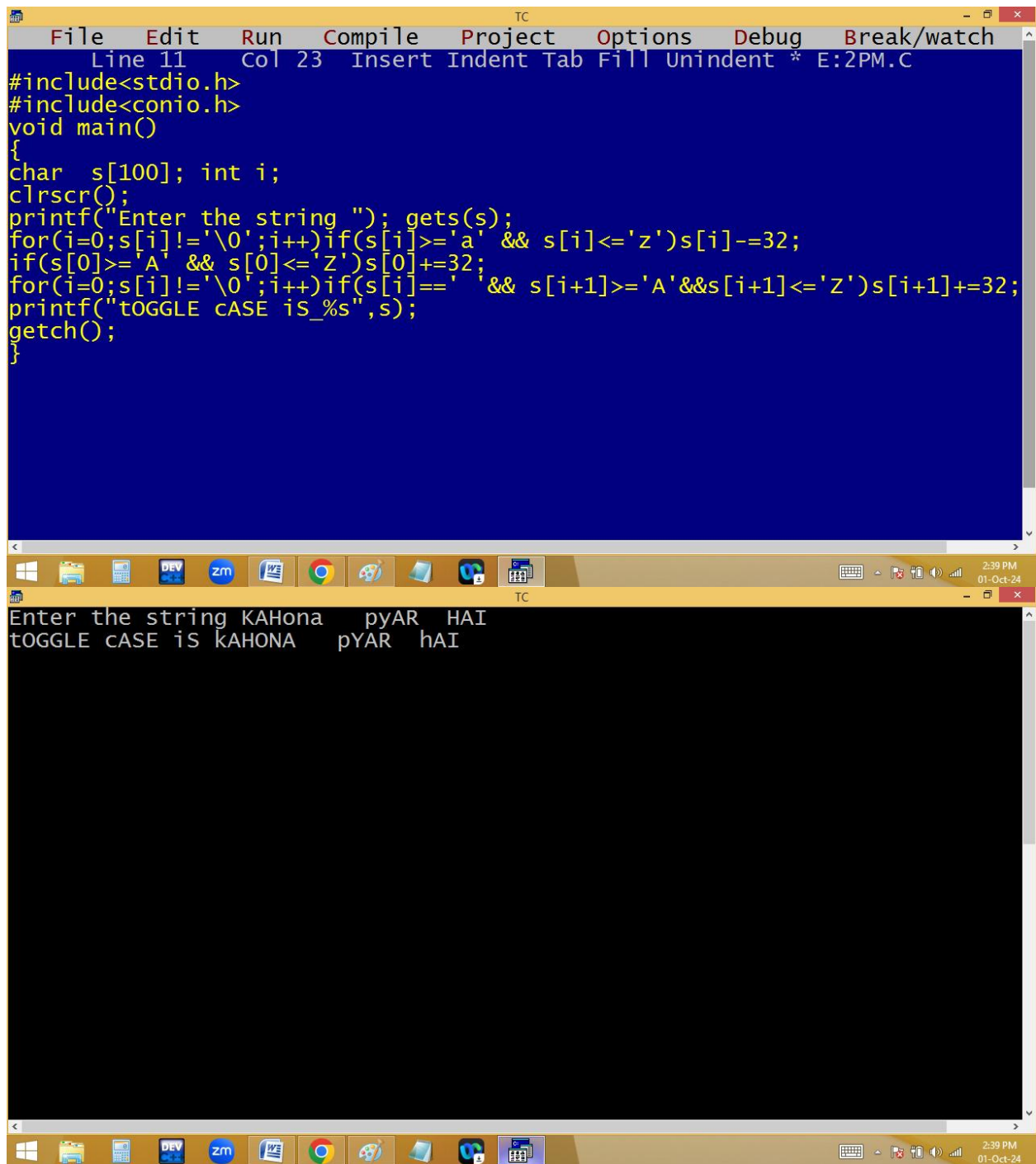
a	b	✓	c	D	✓	e	f	g	h	\0
0	1	2	3	4	5	6	7	8	9	10

```
for(i=0; s[i]!='\0';i++)
if(s[i]>='A' && s[i]<='Z') s[i]+=32; /* strlwr(); */

if(s[0]>='a' && s[0]<='z') s[0]-=32;

for(i=0;s[i]!='\0';i++)
if(s[i]==' ' && s[i+1]>='a' && s[i+1]<='z')s[i+1]-=32
```

tOGGLE cASE:



The image shows a screenshot of a Turbo C++ IDE. The top window displays the source code for a C program named E:2PM.C. The code includes `<stdio.h>` and `<conio.h>`, and defines a `main` function. Inside `main`, it declares a character array `s` of size 100 and an integer `i`. It clears the screen with `clrscr()` and prompts the user to "Enter the string". It then uses `gets(s)` to read the input string. The program iterates through the string, toggling the case of each character: lowercase letters are converted to uppercase by subtracting 32, and uppercase letters are converted to lowercase by adding 32. The modified string is then printed using `printf("tOGGLE cASE iS_%s", s);`, and the program waits for a key press with `getch()` before exiting.

```
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 23 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s[100]; int i;
clrscr();
printf("Enter the string "); gets(s);
for(i=0;s[i]!='\0';i++)if(s[i]>='a' && s[i]<='z')s[i]-=32;
if(s[0]>='A' && s[0]<='Z')s[0]+=32;
for(i=0;s[i]!='\0';i++)if(s[i]==' ' && s[i+1]>='A'&&s[i+1]<='Z')s[i+1]+=32;
printf("tOGGLE cASE iS_%s",s);
getch();
}
```

The bottom window shows the program's execution. The user has entered the string "KAHona pyAR hAI". The output of the program is "tOGGLE CASE iS KAHONA pyAR hAI".

Sentence case:

Kishore working for naresh it. Naresh it. Ameerpet

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 25 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s[100]; int i;
clrscr();
printf("Enter the string "); gets(s);
for(i=0;s[i]!='\0';i++)if(s[i]>='A' && s[i]<='Z')s[i]+=32;
if(s[0]>='a' && s[0]<='z')s[0]-=32;
for(i=0;s[i]!='\0';i++)
if(s[i]=='.' && s[i+1]!=' ' && s[i+2]>='a' && s[i+2]<='z')s[i+2]-=32;
printf("Sentence case is %s",s);
getch();
}
```

Enter the string ram and ra. AMEErpet. hydeRABAD. telENGANA. india.
Sentence case is Ram and ra. Ameerpet. Hyderabad. Telengana. India.

Title Case / Capitalize Each

Word:

s	a	b	✓	c	D	✓	e	F	g	H	\0
	A			c	d		E	f		h	
0	1	2	3	4	5	6	7	8	9	10	

```
for(i=0; s[i]!='\0';i++)
```

```
if(s[i]>='A' && s[i]<='Z') s[i]+=32; /* strlwr(); */
```

```
if(s[0]>='a' && s[0]<='z') s[0]-=32;
```

```
for(i=0;s[i]!='\0';i++)
```

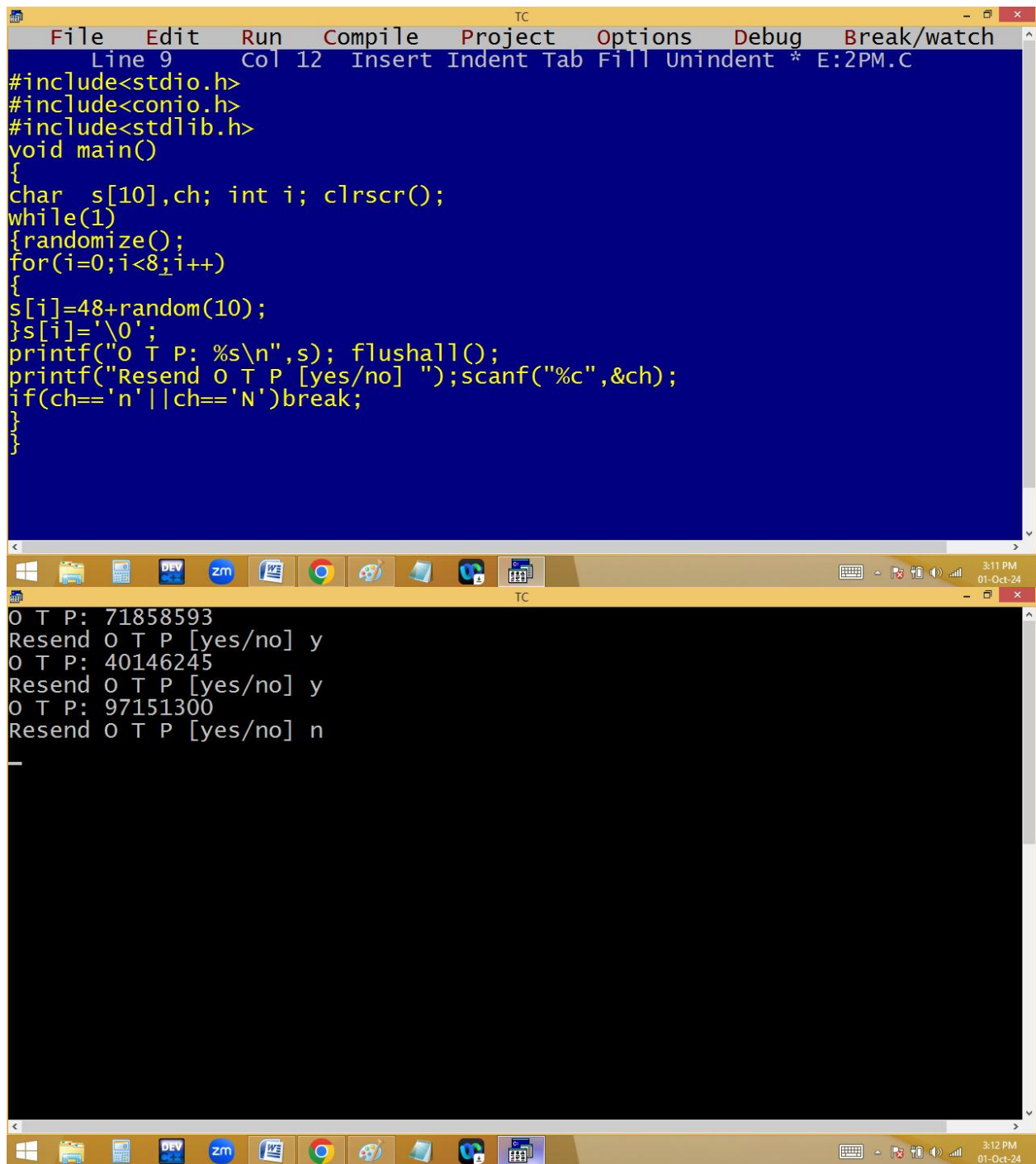
```
if(s[i]==' ' && s[i+1]>='a' && s[i+1]<='z')s[i+1]-=32
```

Captcha code generation:

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 6 Col 40 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
void main()
{
char s[10],sp[]="@#$%&*",ch; int n,i; clrscr();
while(1)
{randomize();
for(i=0;i<6;i++)
{
n=random(4);
if(n==0)s[i]=97+random(26);
else if(n==1)s[i]=65+random(26);
else if(n==2)s[i]=48+random(10);
else s[i]=sp[random(6)];
}s[i]='\0';
printf("Captcha: %s\n",s); fflush();
printf("Refresh Captcha [yes/no] ");scanf("%c",&ch);
if(ch=='n' || ch=='N')break;
}
}
}

Captcha: t@f&v0
Refresh Captcha [yes/no] y
Captcha: E*3RU9
Refresh Captcha [yes/no] y
Captcha: %#y#&h
Refresh Captcha [yes/no] n
```

O T P Generation:



The image shows a screenshot of a Turbo C++ (TC) IDE. The top window displays the source code for a C program named 'E:2PM.C'. The code generates an 8-digit OTP and prompts the user to resend it if they choose 'y'.

```
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 12 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
void main()
{
char s[10],ch; int i; clrscr();
while(1)
{randomize();
for(i=0;i<8;i++)
{
s[i]=48+random(10);
}s[i]='\0';
printf("O T P: %s\n",s); fflush();
printf("Resend O T P [yes/no] ");scanf("%c",&ch);
if(ch=='n' || ch=='N')break;
}
}
```

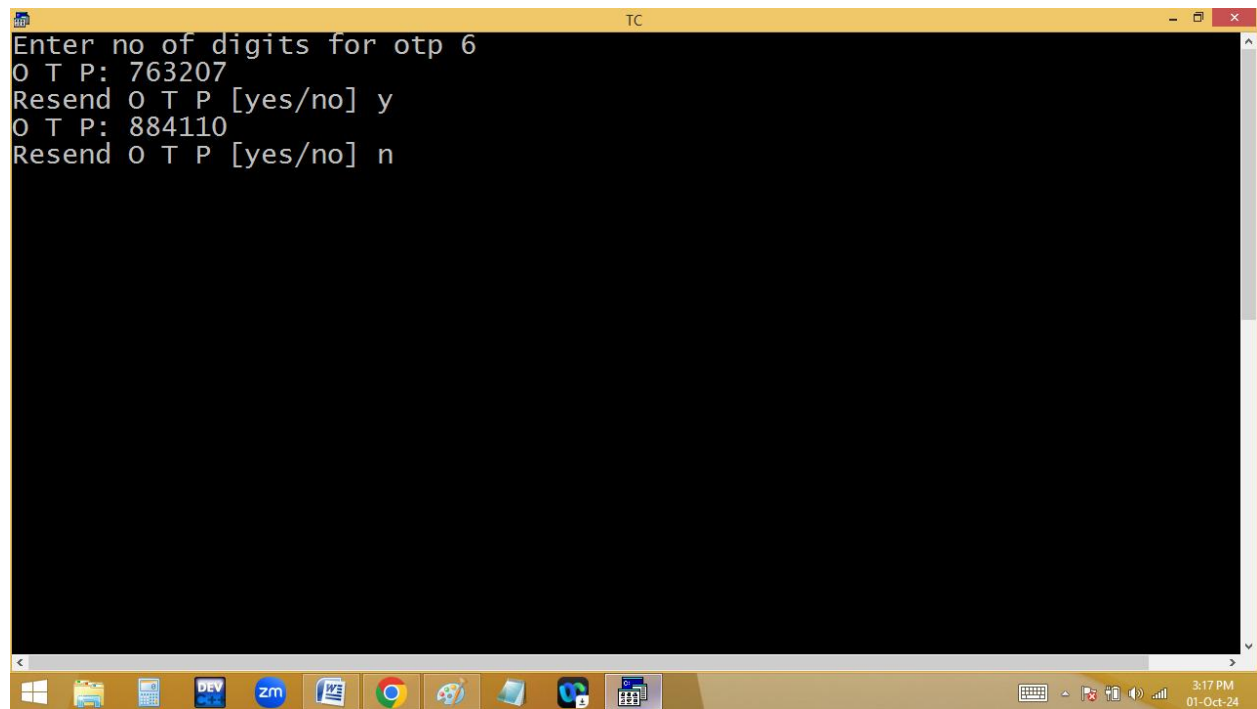
The bottom window shows the program's execution output:

```
O T P: 71858593
Resend O T P [yes/no] y
O T P: 40146245
Resend O T P [yes/no] y
O T P: 97151300
Resend O T P [yes/no] n
```

Otp with dynamic numbering:


```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 23 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h> #include<conio.h> #include<stdlib.h>
void otp(int); /* fun dec */
void otp(int n) /* fun definition */
{
char s[100],ch; int i;
while(1)
{randomize(); for(i=0;i<n;i++)
{ s[i]=48+random(10); }s[i]='\0';
printf("O T P: %s\n",s); fflush();
printf("Resend O T P [yes/no] ");scanf("%c",&ch);
if(ch=='n' || ch=='N')break;
}
}
void main()
{
int n;
clrscr();
printf("Enter no of digits for otp "); scanf("%d",&n);
otp(n); /* fun calling */
}

Enter no of digits for otp 4
O T P: 0687
Resend O T P [yes/no] y
O T P: 1199
Resend O T P [yes/no] y
O T P: 7515
Resend O T P [yes/no] n
```



A screenshot of a Windows 10 desktop environment. A terminal window titled "TC" is open, displaying the following text: "Enter no of digits for otp 6", "O T P: 763207", "Resend O T P [yes/no] y", "O T P: 884110", and "Resend O T P [yes/no] n". The terminal has a black background with white text. The Windows taskbar is visible at the bottom, showing icons for the Start menu, File Explorer, Microsoft Edge, DevTools, Zoom, Word, Chrome, and other applications. The system tray on the right shows the time as 3:17 PM and the date as 01-Oct-24.

```
Enter no of digits for otp 6
O T P: 763207
Resend O T P [yes/no] y
O T P: 884110
Resend O T P [yes/no] n
```

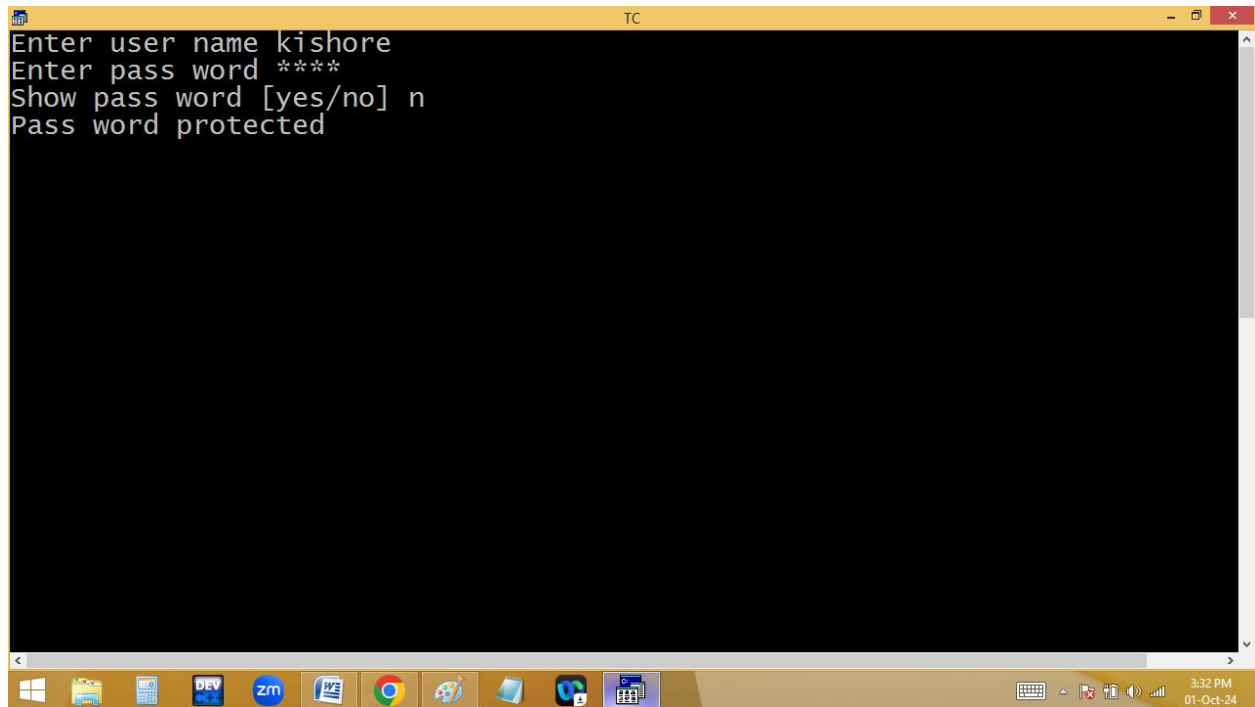
Pass word generation:

The image shows two screenshots of the Turbo C++ (TC) IDE. The top screenshot displays the source code of a C program designed to protect a password. The code includes headers for `stdio.h` and `conio.h`, and defines a `main` function. It declares character arrays for `user` and `pass`, and an integer `i`. The program prompts the user for a username and password, then enters a loop where it prompts for a confirmation to show the password. If the user confirms, it prints the password; otherwise, it prints "Pass word protected".

```
File Edit Run Compile Project Options Debug Break/watch
Line 18 Col 9 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char user[20], pass[20],ch; int i=0;
clrscr();
printf("Enter user name "); scanf("%s",user);
printf("Enter pass word ");
while((ch=getch())!=13)
{
printf("*");
pass[i]=ch;
i++;
} pass[i]='\0';
printf("\nShow pass word [yes/no] ");flushall();scanf("%c",&ch);
if(ch=='y' || ch=='Y') printf("Ur pass word %s",pass);
else puts("Pass word protected");
getch();
}
```

The bottom screenshot shows the program's execution. The user has entered "Kishore" for the username, "*****" for the password, and "y" for the confirmation. The program has printed "Ur pass word jackson_".

```
Enter user name Kishore
Enter pass word *****
Show pass word [yes/no] y
Ur pass word jackson_
```



```
Enter user name kishore
Enter pass word ****
Show pass word [yes/no] n
Pass word protected
```

STRING LIBRARY FUNCTIONS

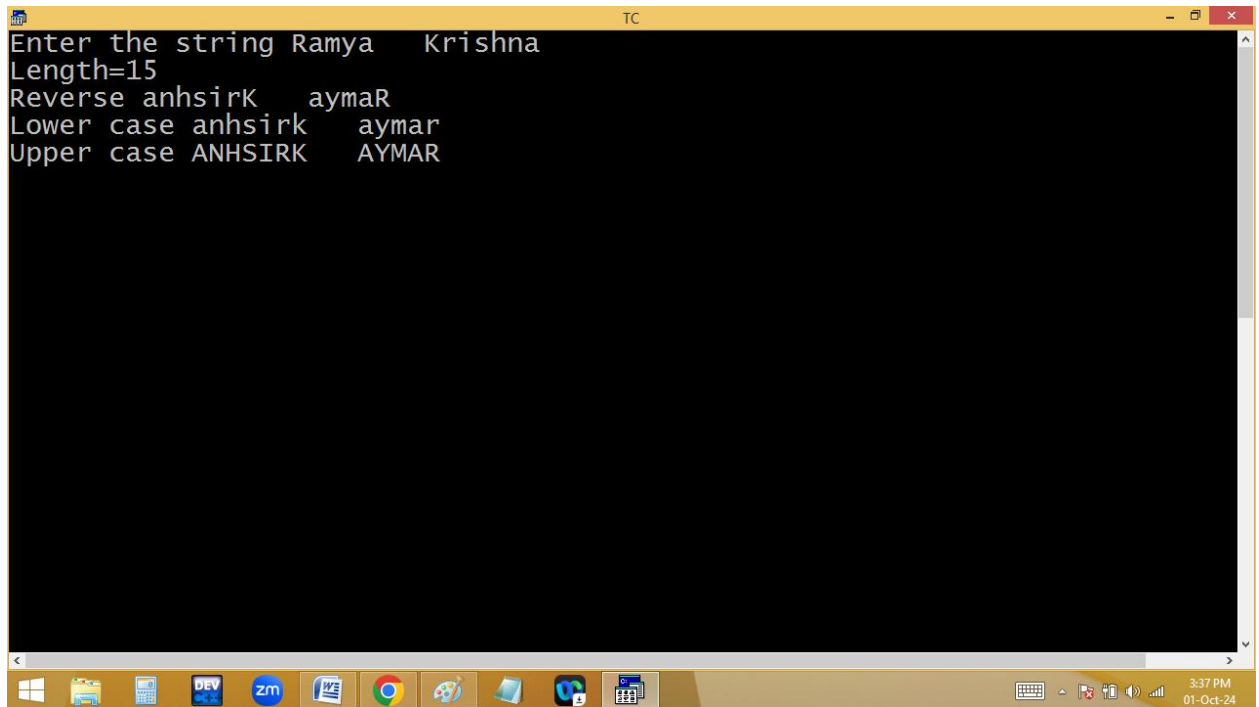
To manage string operations c language provides some predefined functions available in <string.h>

1. Strlen(): Return string length
2. Strrev(): Returns reverse string
3. Strlwr(): Converts into lower case.
4. Strupr(): converts into upper case.

The image shows two screenshots of the Turbo C++ (TC) IDE. The top screenshot displays the source code for a C program that processes a string. The code includes headers for `stdio.h`, `conio.h`, and `string.h`. The `main` function prompts the user to enter a string, then prints its length, reverse, lower case, and upper case versions. The bottom screenshot shows the program's execution output for the input string "RamU".

```
File Edit Run Compile Project Options Debug Break/watch
Line 14 Col 2 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
#include<string.h>
void main()
{
char s[100];
clrscr();
printf("Enter the string "); gets(s);
printf("Length=%d\n",strlen(s));
printf("Reverse %s\n",strrev(s));
printf("Lower case %s\n",strlwr(s));
printf("Upper case %s",strupr(s));
getch();
}
```

Enter the string RamU
Length=4
Reverse UmaR
Lower case umar
Upper case UMAR



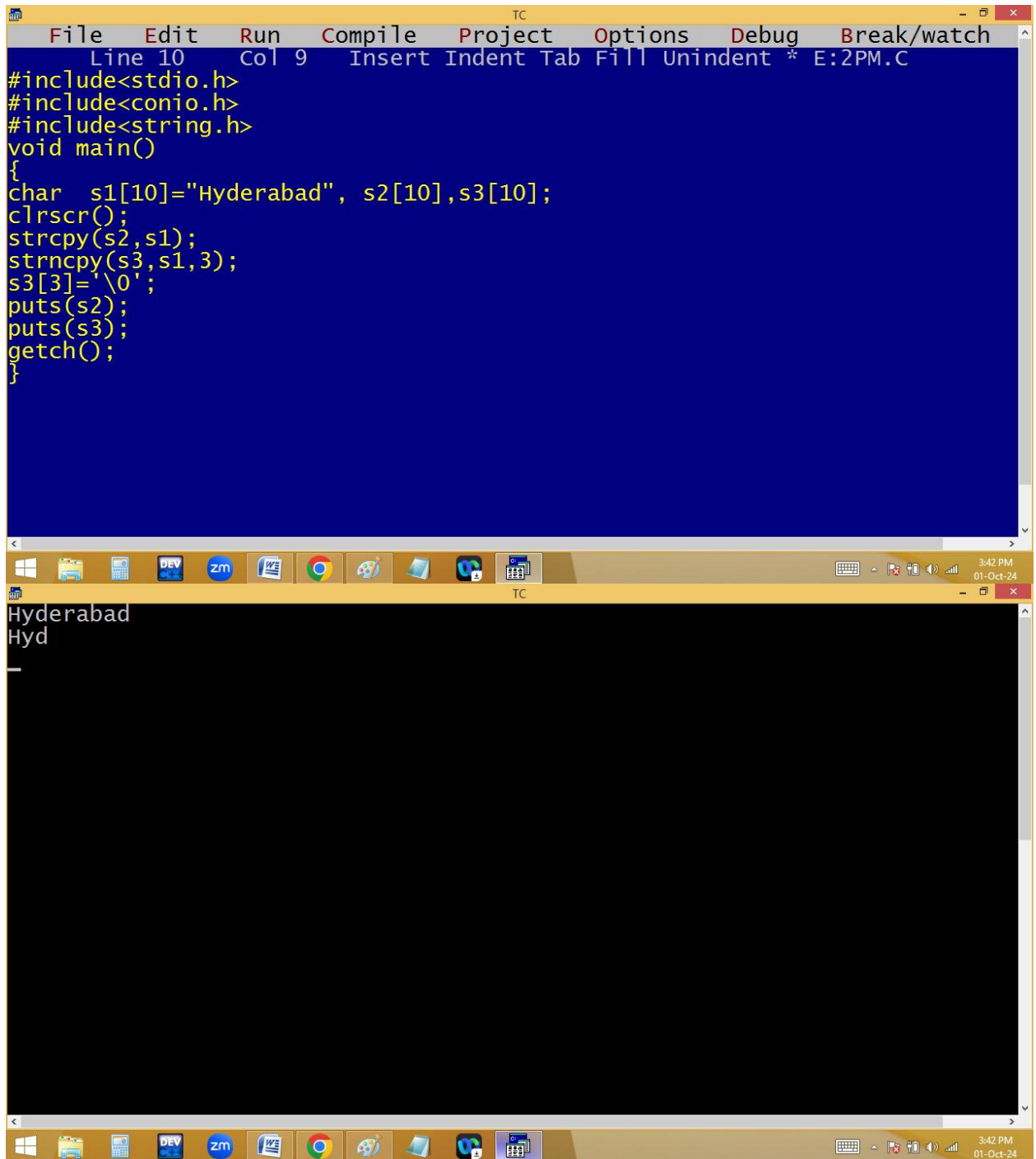
```
Enter the string Rama Krishna
Length=15
Reverse anhsirk aymaR
Lower case anhsirk aymar
Upper case ANHSIRK AYMAR
```

5. Strcpy(): It copies source string into destination string.

Strcpy(destination string, source string);

6. Strncpy(): It copies specified no of char's into destination string.

Strncpy(dest string, source string, no of char's);



The screenshot shows a Turbo C++ IDE window titled 'TC'. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the bottom indicates 'Line 10', 'Col 9', and 'Insert Indent Tab Fill Unindent * E:2PM.C'. The code editor contains the following C program:

```
#include<stdio.h>
#include<conio.h>
#include<string.h>
void main()
{
char s1[10]="Hyderabad", s2[10],s3[10];
clrscr();
strcpy(s2,s1);
strncpy(s3,s1,3);
s3[3]='\0';
puts(s2);
puts(s3);
getch();
}
```

Below the code editor is a console window titled 'TC' showing the output of the program:

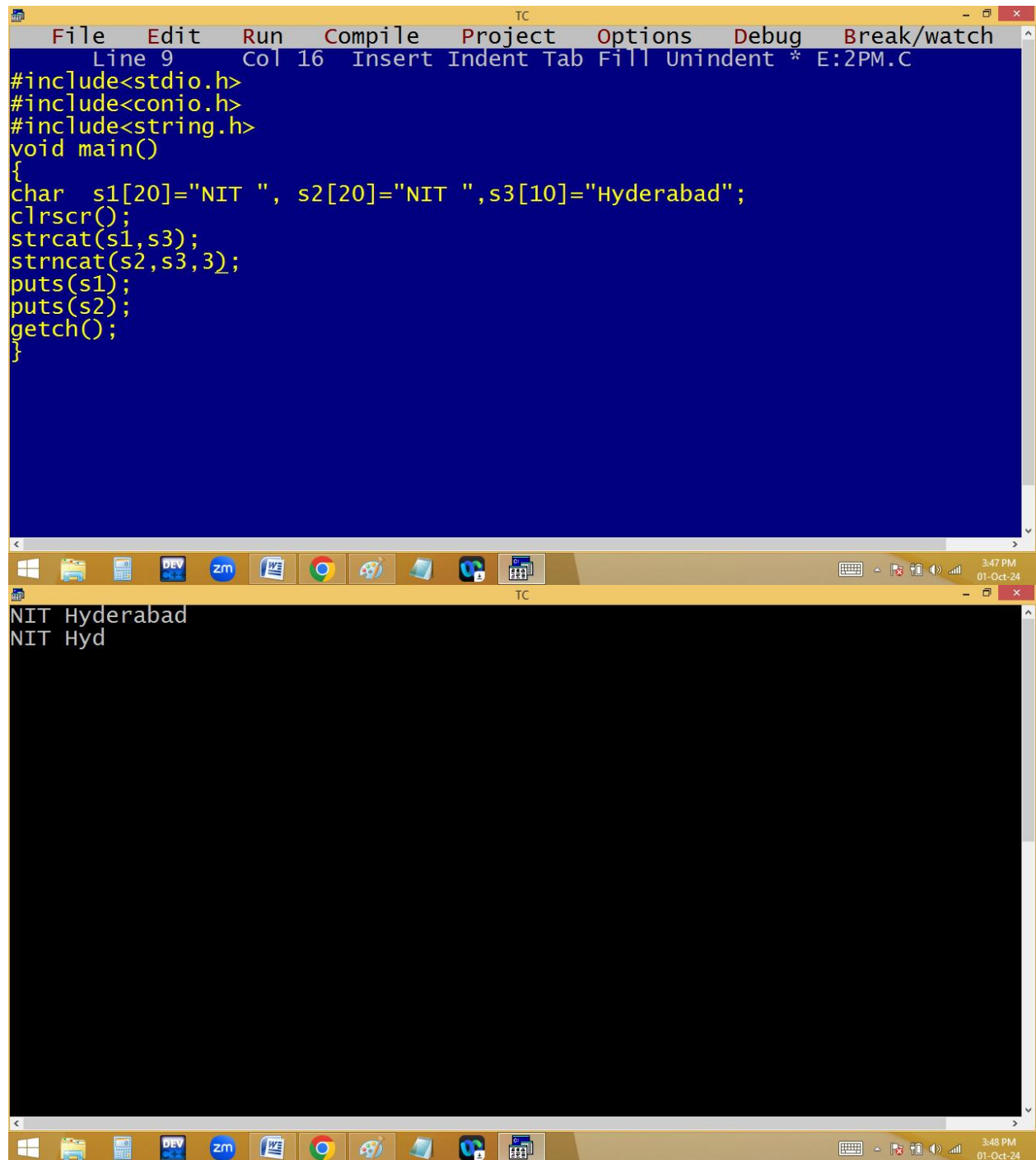
```
Hyderabad
Hyd
_
```

7. Strcat(): It adds second string to first string.

Strcat(string1, string2);

8. Strncat(): It adds specified no of char's to string1.

Strncat(string1, string2, no of characters);



The image shows two screenshots of the Turbo C++ (TC) IDE. The top screenshot displays the source code of a C program. The code includes headers for `stdio.h`, `conio.h`, and `string.h`. It defines a `main` function where three character arrays are declared: `s1[20]` with the value "NIT ", `s2[20]` with the value "NIT ", and `s3[10]` with the value "Hyderabad". The program then calls `clrscr()` to clear the screen, followed by `strcat(s1, s3);` to concatenate the entire string "Hyderabad" to the end of `s1`. Next, it calls `strncat(s2, s3, 3);` to concatenate only the first 3 characters of `s3` ("Hyd") to the end of `s2`. Finally, it prints `s1` and `s2` using `puts()` and waits for a key press with `getch()`. The bottom screenshot shows the output of the program. The first line of output is "NIT Hyderabad" and the second line is "NIT Hyd". The Windows taskbar at the bottom of both screenshots shows the time as 3:47 PM and 3:48 PM on 01-Oct-24.

```
#include<stdio.h>
#include<conio.h>
#include<string.h>
void main()
{
char  s1[20]="NIT ", s2[20]="NIT ",s3[10]="Hyderabad";
clrscr();
strcat(s1,s3);
strncat(s2,s3,3);
puts(s1);
puts(s2);
getch();
}
```

NIT Hyderabad
NIT Hyd